

2012
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 17, 19 and 23

Total marks – 100

Section I Pages 2–25

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–33
- Allow about 1 hour and 40 minutes for this part

Section II Pages 27–38

25 marks

- Attempt ONE question from Questions 34–38
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

1 Which of the following is a measure of the clarity of water?

- (A) Hardness
- (B) Turbidity
- (C) Total dissolved solids
- (D) Biochemical oxygen demand



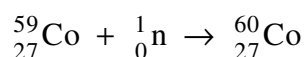
Which of the following compounds is represented by X in the flowchart?

- (A) Cellulose
 - (B) Ethanol
 - (C) Glucose
 - (D) Styrene
- 3 What effect does a catalyst have on a reaction?
- (A) It increases the rate.
 - (B) It increases the yield.
 - (C) It increases the heat of reaction.
 - (D) It increases the activation energy.
- 4 Which pieces of glassware should be used when preparing a primary standard solution?
- (A) Pipette, burette and conical flask
 - (B) Dropper, watch glass and pipette
 - (C) Beaker, filter funnel and volumetric flask
 - (D) Measuring cylinder, stirring rod and conical flask

5 Which of the following is a balanced equation representing the fermentation of glucose?

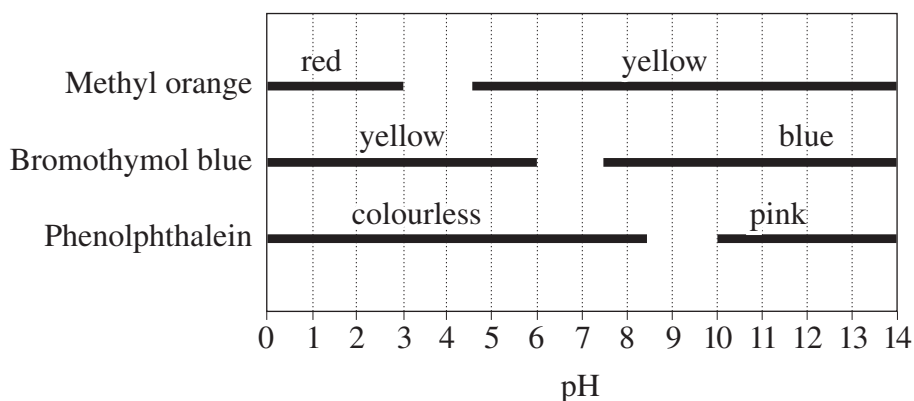
- (A) $\text{C}_6\text{H}_{12}\text{O}_6(aq) \rightarrow 2\text{C}_3\text{H}_6\text{O}_3(aq)$
(B) $\text{C}_6\text{H}_{12}\text{O}_6(aq) \rightarrow 2\text{C}_2\text{H}_5\text{OH}(aq) + 2\text{CO}_2(g)$
(C) $\text{C}_6\text{H}_{12}\text{O}_6(aq) + 6\text{O}_2(g) \rightarrow 6\text{CO}_2(g) + 6\text{H}_2\text{O}(l)$
(D) $\text{C}_6\text{H}_{12}\text{O}_6(aq) + 3\text{O}_2(g) \rightarrow \text{C}_2\text{H}_5\text{OH}(aq) + 4\text{CO}_2(g) + 3\text{H}_2\text{O}(l)$

6 Cobalt-60 is produced according to the equation:



Where would a commercial quantity of cobalt-60 be produced?

- (A) Cyclotron
(B) Scintillator
(C) Nuclear reactor
(D) Particle accelerator
- 7 Methyl orange, bromothymol blue and phenolphthalein indicators were mixed together to form a solution.



Over what pH range would the solution be yellow?

- (A) 0 – 14
(B) 3 – 4.5
(C) 3 – 7.5
(D) 4.5 – 6

8 Which acid / base pair could act as a buffer?

- (A) $\text{H}_3\text{O}^+ / \text{H}_2\text{O}$
- (B) $\text{H}_2\text{O} / \text{OH}^-$
- (C) $\text{HNO}_3 / \text{NO}_3^-$
- (D) $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$

9 Which of the following contains a coordinate covalent bond?

- (A) NH_3
- (B) NH_4^+
- (C) H_2O
- (D) OH^-

10 Samples of a solution of barium nitrate were independently tested with chloride ions, with sulfate ions and also for flame colour.

Which row of the following table would represent the results?

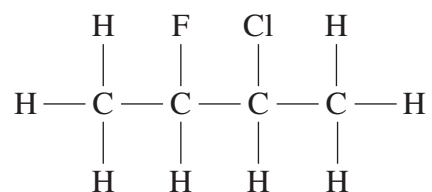
	<i>Chloride</i>	<i>Sulfate</i>	<i>Flame test</i>
(A)	No precipitate	No precipitate	Red
(B)	No precipitate	Precipitate	Green
(C)	Precipitate	Precipitate	Green
(D)	Precipitate	No precipitate	Red

11 The pH of 0.1 mol L^{-1} solutions of acetic, citric and hydrochloric acids was measured.

Which solution has the highest pH?

- (A) Citric acid
- (B) Acetic acid
- (C) Hydrochloric acid
- (D) The pH of the three solutions is the same.

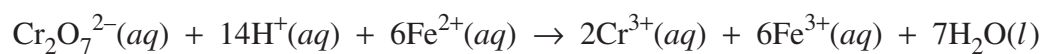
12 What is the correct IUPAC name for the following compound?



- (A) 2-chloro-2-fluorobutane
- (B) 2-fluoro-3-chlorobutane
- (C) 3-fluoro-2-chlorobutane
- (D) 3-chloro-2-fluorobutane

Use the information provided to answer Questions 13 and 14.

This equation represents a common redox reaction.



13 What is the oxidising agent in the reaction?

- (A) H^+
- (B) Cr^{3+}
- (C) Fe^{2+}
- (D) $\text{Cr}_2\text{O}_7^{2-}$

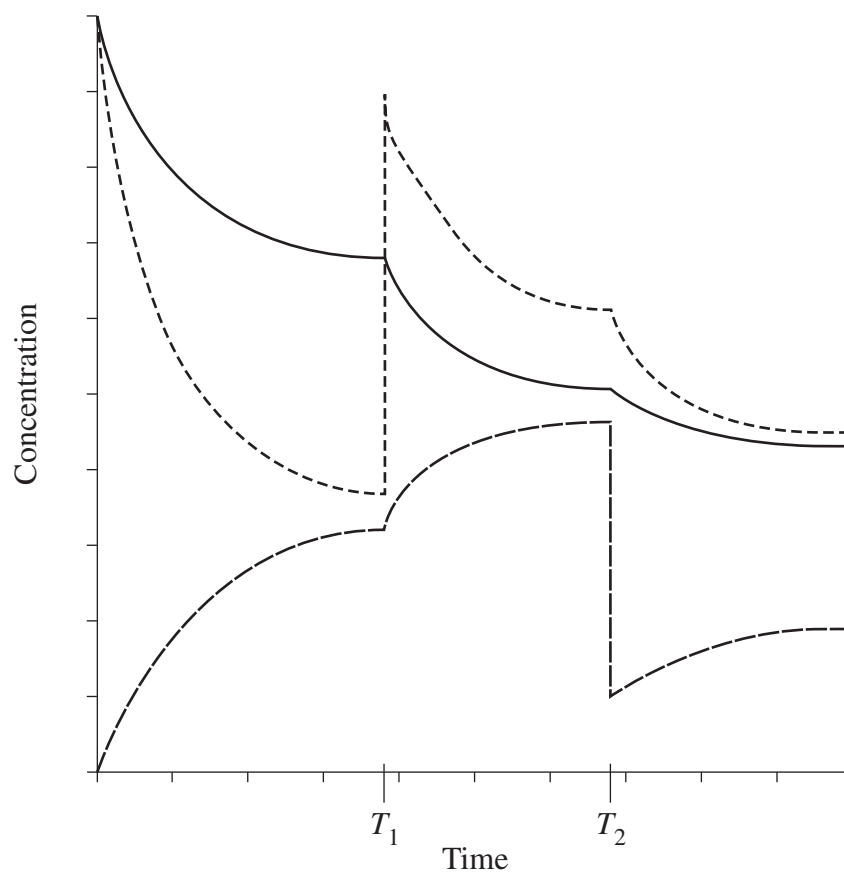
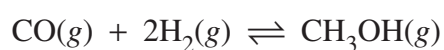
14 What is the value of $E_{\text{cell}}^\ominus$ for the reaction?

- (A) 0.59 V
- (B) 0.92 V
- (C) 1.90 V
- (D) 2.13 V

15 In which row of the following table are the listed oxides correctly classified?

	<i>Acidic</i>	<i>Basic</i>	<i>Neutral</i>	<i>Amphoteric</i>
(A)	CO ₂	Na ₂ O	SO ₃	Al ₂ O ₃
(B)	Na ₂ O	CO ₂	H ₂ O	Al ₂ O ₃
(C)	CO ₂	MgO	H ₂ O	ZnO
(D)	SO ₂	K ₂ O	CO	CO ₂

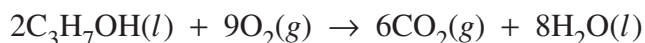
16 The graph shows the concentrations over time for the system:



What has happened at times T_1 and T_2 ?

	T_1	T_2
(A)	H ₂ added	CH ₃ OH removed
(B)	CO added	CH ₃ OH removed
(C)	H ₂ added	CO removed
(D)	CO added	CO and H ₂ removed

- 17 The heat of combustion of propan-1-ol is 2021 kJ mol^{-1} . Combustion takes place according to the equation:



What mass of water is formed when 1530 kJ of energy is released?

- (A) 3.4 g
(B) 14 g
(C) 55 g
(D) 144 g
- 18 Which of the following changes take place when 50 mL of water is added to 50 mL of 0.1 mol L^{-1} acetic acid?

	<i>pH</i>	<i>Degree of ionisation</i>
(A)	Increase	Decrease
(B)	Decrease	Increase
(C)	Increase	Increase
(D)	Decrease	Decrease

- 19 What mass of anhydrous sodium carbonate is required to neutralise 100.0 mL of 0.500 mol L^{-1} acetic acid?

- (A) 2.65 g
(B) 5.30 g
(C) 10.6 g
(D) 53.0 g

- 20 All the lead ions present in a 50.0 mL solution were precipitated by reaction with excess chloride ions. The mass of the dried precipitate was 0.595 g.

What was the concentration of lead in the original solution?

- (A) 8.87 g L^{-1}
(B) 10.2 g L^{-1}
(C) 11.9 g L^{-1}
(D) 16.0 g L^{-1}

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Chemistry

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks

Attempt Questions 21–33

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (4 marks)

- (a) Write a balanced chemical equation, using structural formulae, for the formation of ethyl butanoate. 2

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- (b) Common safety precautions in school laboratories include the use of safety glasses, gloves and lab coats. Justify the use of another safety precaution specifically required to safely make ethyl butanoate. 2

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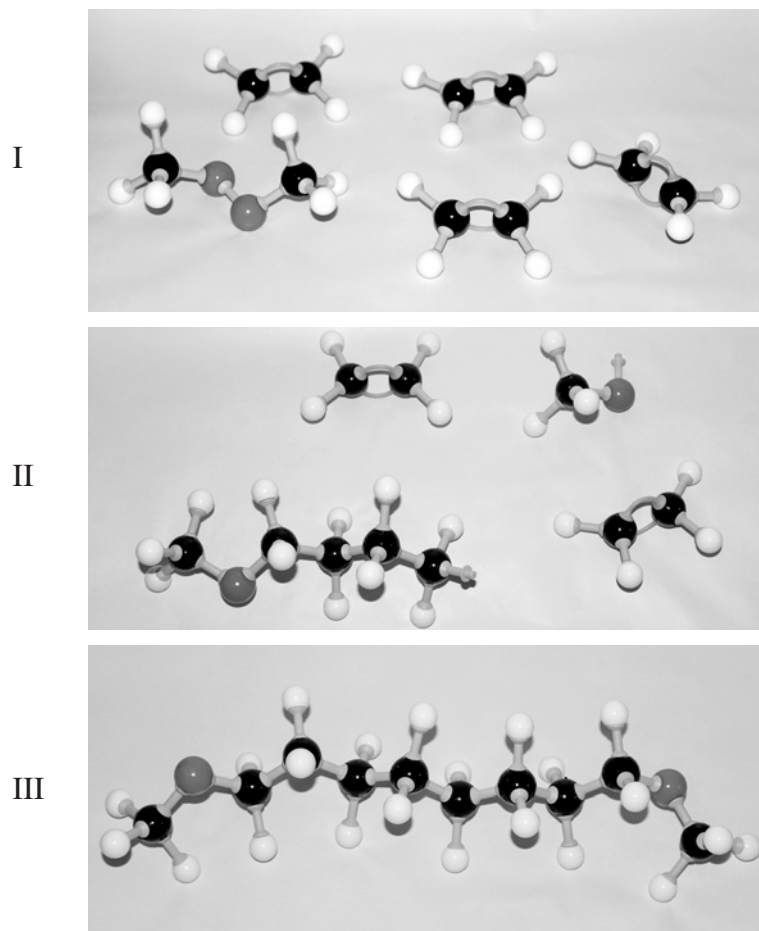
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Question 22 (3 marks)

A student created the following models to demonstrate a chemical process.



(a) What is the chemical process being modelled?

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(b) Why are models such as these useful?

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Centre Number

Section I – Part B (continued)

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Student Number

Question 23 (3 marks)

Explain the impact of an increase in pressure and an increase in temperature on the solubility of carbon dioxide in water. Include a relevant equation in your answer.

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Question 24 (3 marks)

Explain why ammonia is such an important raw material in industry today.

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Centre Number

Section I – Part B (continued)

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Question 25 (3 marks)

Describe the process of monitoring waterways for eutrophication.

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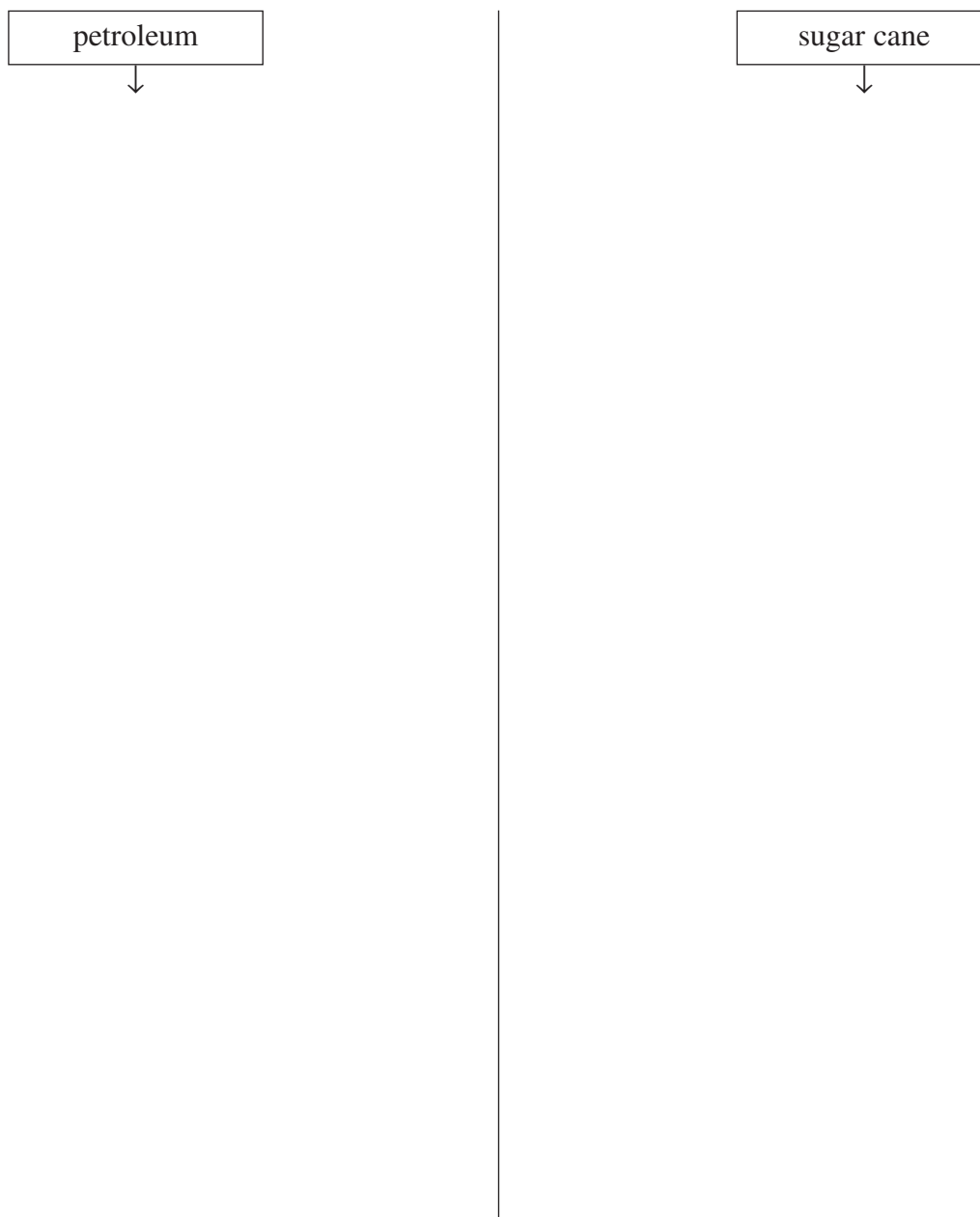
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Question 26 (8 marks)

Petroleum and sugar cane are both raw materials used for the production of ethanol.

- (a) Construct separate flow diagrams for the production of ethanol from each raw material.

5



Question 26 continues on page 15

Question 26 (continued)

- (b) Compare the environmental sustainability of producing ethanol from these two raw materials.

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End of Question 26

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Centre Number

Section I – Part B (continued)

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Student Number

Question 27 (3 marks)

Iodine-131 decays through both beta and gamma emission. Iodine-123 decays through gamma emission only.

- (a) Iodine-131 is used for diagnosis and therapy whereas Iodine-123 is used only for diagnosis. 2

	beta emission	gamma emission
Emitted particle	electron	gamma-ray
Ability to pass through biological tissue	low	high

With reference to the information and the table, justify the different uses of these two radioisotopes.

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- (b) Write the equation representing the decay of Iodine-131 by beta emission. 1

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Question 28 (3 marks)

A solution was made by mixing 75.00 mL of 0.120 mol L^{-1} hydrochloric acid with 25.00 mL of 0.200 mol L^{-1} sodium hydroxide.

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What is the pH of the solution?

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Centre Number

Section I – Part B (continued)

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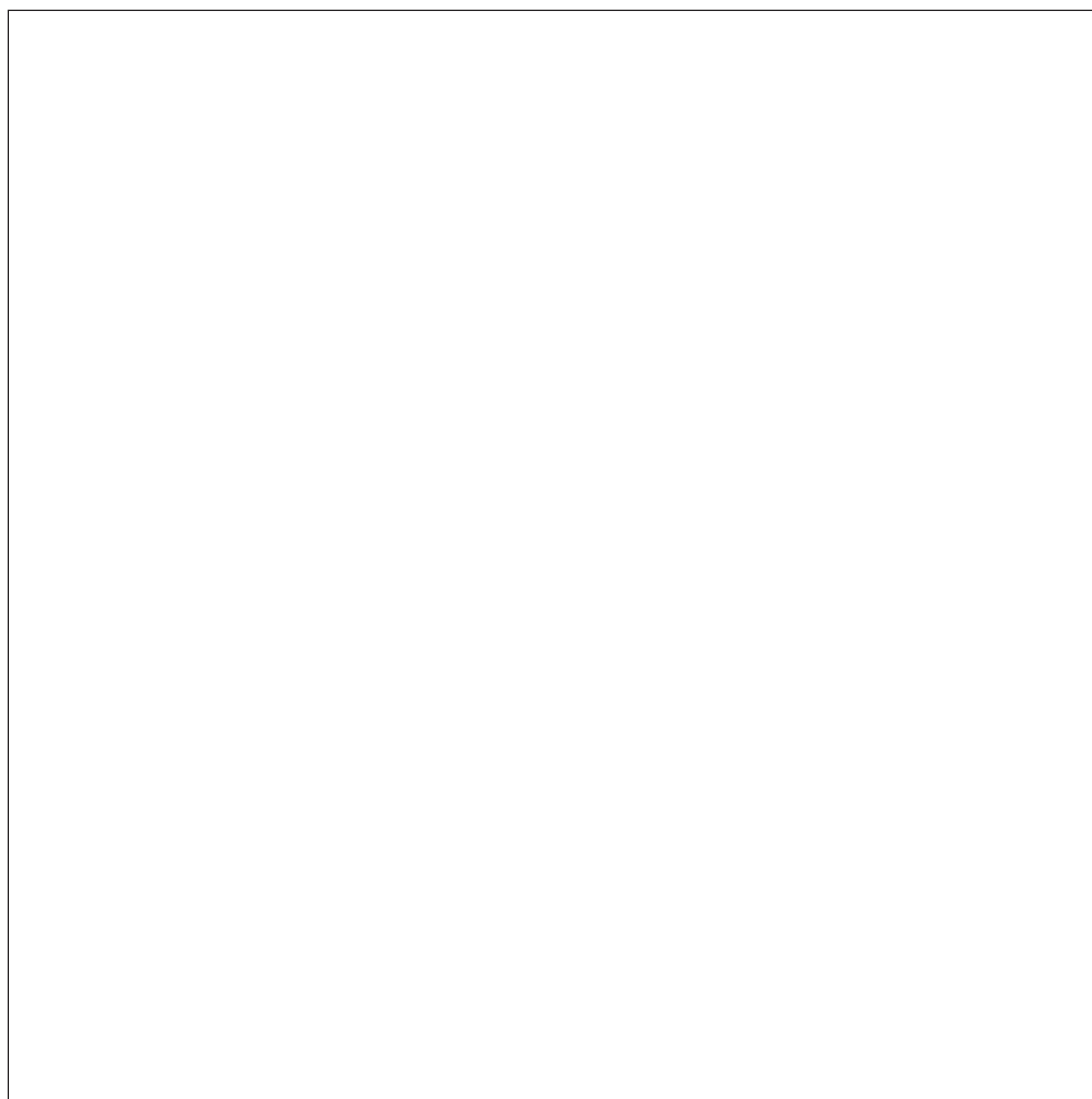
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Question 29 (5 marks)

Draw a labelled diagram to show the layered structure of the atmosphere. In your diagram include:

5

- the names of TWO atmospheric pollutants, positioned in the layers where the detrimental impact occurs
- the names of the sources of the two pollutants identified.



Question 30 (6 marks)

A chemist analysed aspirin tablets for quality control. The initial step of the analysis was the standardisation of a NaOH solution. Three 25.00 mL samples of a 0.1034 mol L⁻¹ solution of standardised HCl were titrated with the NaOH solution. The average volume required for neutralisation was 25.75 mL.

- (a) Calculate the molarity of the NaOH solution.

2

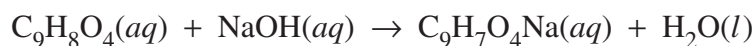
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Three flasks were prepared each containing a mixture of 25 mL of water and 10 mL of ethanol. An aspirin tablet was dissolved in each flask. The aspirin in each solution was titrated with the standardised NaOH solution according to the following equation:



The following titration results were obtained.

<i>Tablet</i>	<i>Volume (mL)</i>
1	16.60
2	16.50
3	16.55

- (b) (i) Calculate the average mass (mg) of aspirin per tablet.

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- (ii) Why was it necessary to include the ethanol in the mixture?

1

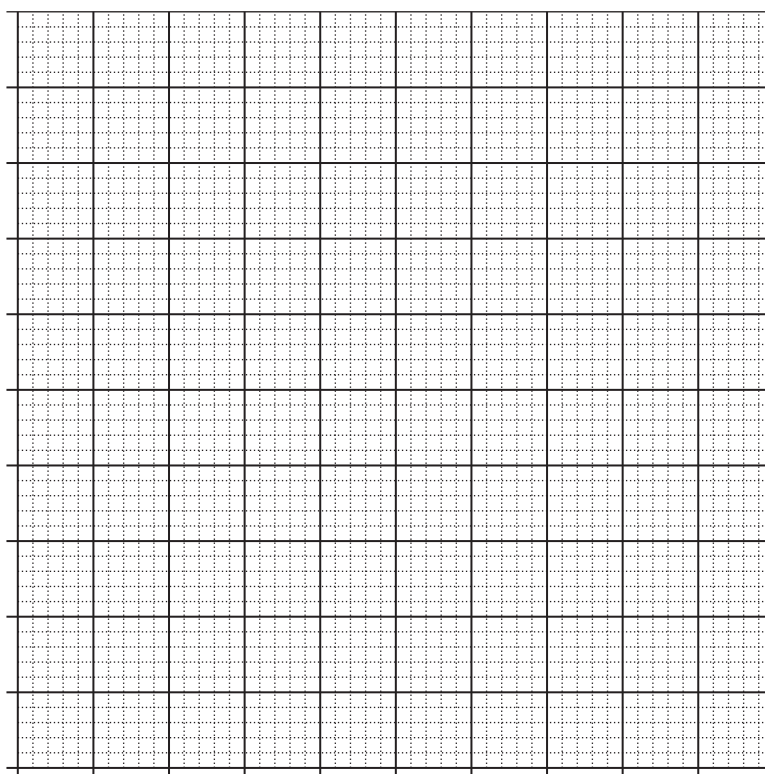
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Question 31 (5 marks)

The boiling points of some alkanols are given in the table.

<i>Alkanol</i>	<i>Boiling point (°C)</i>
Methanol	65
Ethanol	79
Propan-1-ol	97
Pentan-1-ol	138
Hexan-1-ol	157
Heptan-1-ol	176

- (a) Using the data provided, construct a graph that shows the relationship between carbon chain length and boiling point.

3

- (b) Using the graph, predict the boiling point of butan-1-ol.

1

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- (c) What is the intermolecular force responsible for the trend shown in the graph?

1

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Centre Number

Section I – Part B (continued)

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Student Number

Question 32 (3 marks)

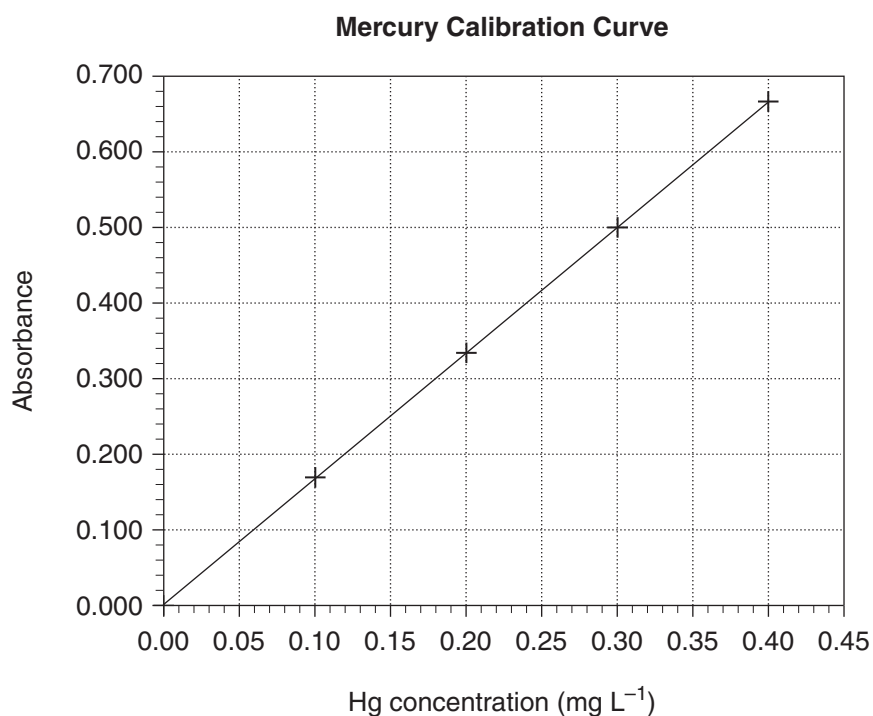
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Question 32 (3 marks)

The mercury concentration of a certain fish species was determined by atomic absorption spectroscopy. The sample data are:

3

Mass of fish (g)	18.6
Final sample volume (mL)	25.0
Absorbance (mean)	0.280



A consumer wants to avoid eating fish with a mercury concentration greater than 0.5 mg/kg of fish.

Calculate the concentration of mercury in the fish sample and state whether the consumer can eat this fish species.

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Question 33 (6 marks)

Chemists can assist in reversing or minimising the environmental problems caused by technology and the human demand for products and services.

6

With reference to this statement, assess the need for chemists to collaborate when monitoring the environmental impact of a named electrochemical cell.

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Chemistry

Section II

25 marks**Attempt ONE question from Questions 34–38****Allow about 45 minutes for this section**

Answer parts (a)–(c) of the question in Section II Answer Booklet 1.

Answer parts (d)–(e) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

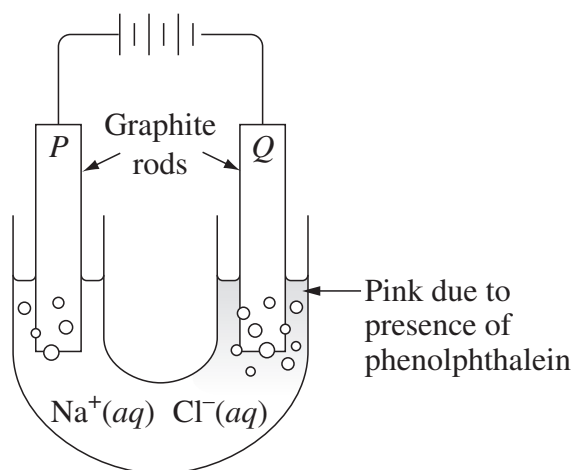
Show all relevant working in questions involving calculations.

	Pages
Question 34 Industrial Chemistry	28–29
Question 35 Shipwrecks, Corrosion and Conservation	30–31
Question 36 The Biochemistry of Movement	32–33
Question 37 The Chemistry of Art	34–35
Question 38 Forensic Chemistry	36–38

Question 34 — Industrial Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The following equipment was set up and the reaction allowed to proceed. Gases were produced at both electrodes. **3**



Name this process and identify the gas at each electrode.

- (b) The equilibrium constant expression for a gaseous reaction is as follows:

$$K = \frac{[\text{N}_2][\text{O}_2]}{[\text{NO}]^2}$$

- (i) Write the equation for this reaction. **1**
- (ii) 0.400 moles of NO was placed in a 1.00 L vessel at 2000°C. The equilibrium concentration of N₂ was found to be 0.198 mol L⁻¹. **3**

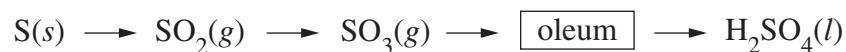
Calculate the equilibrium constant for this reaction and use this value to describe the position of the equilibrium.

- (iii) What could be changed that would result in a different value of K for this equilibrium? **1**

Question 34 continues on page 29

Question 34 (continued)

(c) The production of sulfuric acid is shown.



(i) Describe the production of oleum and its conversion to concentrated sulfuric acid. Include chemical equations in your answer. **3**

(ii) SO_3 can react with water to produce a solution of H_2SO_4 . **2**

Why is it essential to convert SO_3 to oleum before the formation of H_2SO_4 ?

Answer parts (d)–(e) in Section II Answer Booklet 2.

(d) (i) Outline how one of the steps involved in the Solvay process can be chemically modelled in the school laboratory. Include a balanced chemical equation in your answer. **3**

(ii) Identify ONE risk factor and ONE difficulty associated with the laboratory modelling of the step. **2**

(e) Initially soap was the only product of the surfactant industry. Due to societal pressures and chemical developments, production in this industry has evolved to include a wide range of products. **7**

Account for these changes over time with reference to the structure and uses of surfactants.

End of Question 34

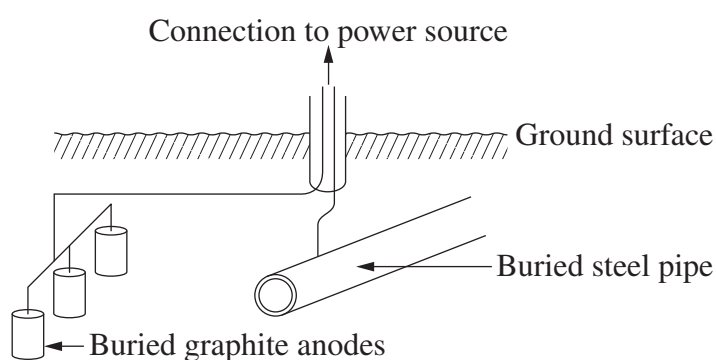
Question 35 — Shipwrecks, Corrosion and Conservation (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) A sealed container transporting a recently recovered artefact is damaged, allowing seawater to escape while appearing to leave the artefact intact. **3**

Why would the loss of seawater be of concern to the maritime archaeologists receiving the artefact?

- (b) The diagram illustrates one method of protecting a steel pipe.

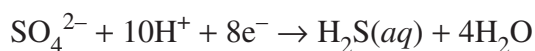


- (i) Explain how this method works. **3**
- (ii) Suggest an alternative way the pipe can be chemically protected. Use half-equations to support your answer. **2**

Question 35 continues on page 31

Question 35 (continued)

- (c) Corrosion in some wrecks can be represented in part by the following half-equation.

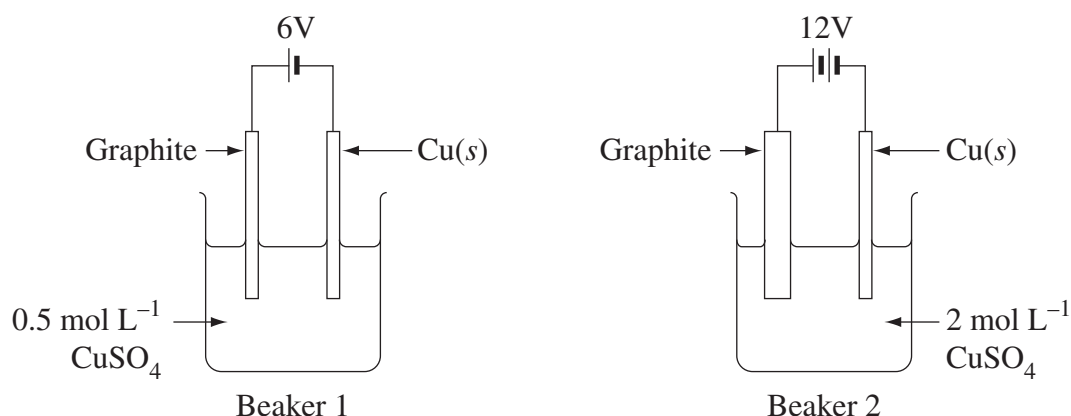


- (i) Under what conditions would this process occur naturally? 2
- (ii) These conditions were simulated in a laboratory by placing a piece of steel in an appropriate solution. The amount of hydrogen sulfide produced was monitored over time. 3

Calculate the loss of iron, in grams, from the piece of steel if 0.76 g of hydrogen sulfide was produced. Include a balanced equation in your answer.

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A student set up the following two beakers to investigate factors affecting electrolysis.



- (i) Why is this investigation not valid? 1
- (ii) Draw diagrams to show how this investigation should be carried out to accurately identify TWO factors affecting electrolysis. Indicate the expected results on your diagrams. 4
- (e) Critically evaluate the continued use of steel in ship building with reference to the history of ocean-going vessels, the composition of steel and ways in which its interaction with the environment can be controlled. 7

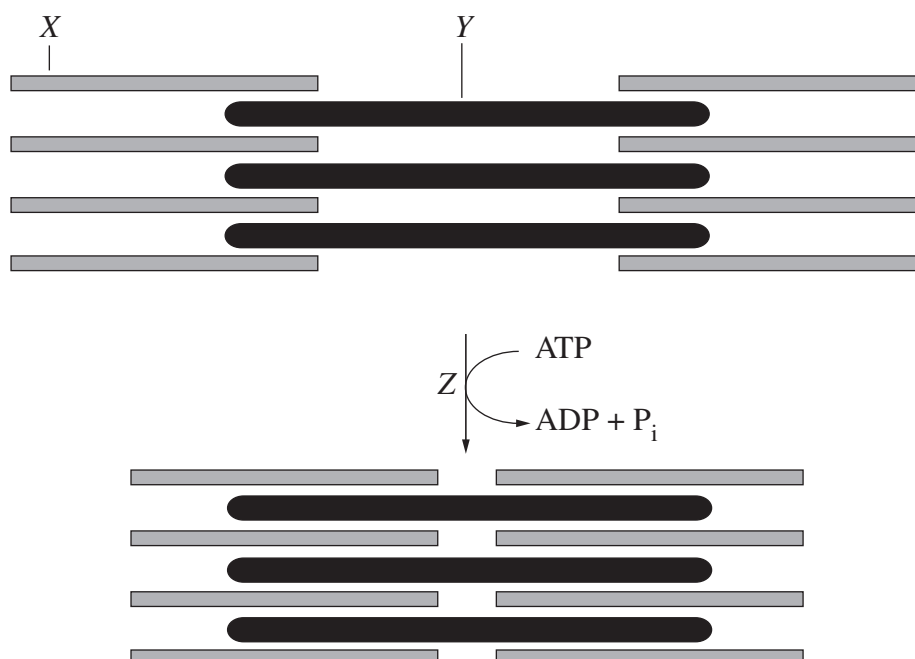
End of Question 35

Question 36 — The Biochemistry of Movement (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The diagram illustrates schematically the current model of muscle contraction.

3



Name protein X, protein Y and metal ion Z.

- (b) (i) Provide the general formula for a fatty acid. **1**
- (ii) Provide the structural formula for glycerol. **1**
- (iii) Account for the difference in the aqueous solubility of glycerol compared to that of fatty acids and triacylglycerols (TAGs). **3**

Question 36 continues on page 33

Question 36 (continued)

- (c) (i) Why do fats have a greater energy density than carbohydrates? **1**
- (ii) Carbohydrates and fats have parts of their oxidative metabolic pathways in common. **4**

Use a flow diagram to outline the main stages of the aerobic metabolism of fats and carbohydrates.

In your diagram clearly identify:

- separate pathways
- common pathway
- stage where most of the ATP is produced
- stage where most of the carbon dioxide is released.

Answer parts (d)–(e) in Section II Answer Booklet 2.

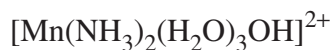
- (d) (i) Compare the forces and bonds that determine the level of structure in proteins. **3**
- (ii) How is the secondary structure of proteins affected by changes in pH? **2**
- (e) Compare and contrast the aerobic and anaerobic metabolism of glucose within the context of muscle function during different forms of exercise and the relative amounts of energy released. **7**

End of Question 36

Question 37 — The Chemistry of Art (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

(a) The formulae of three coordination complexes are shown.

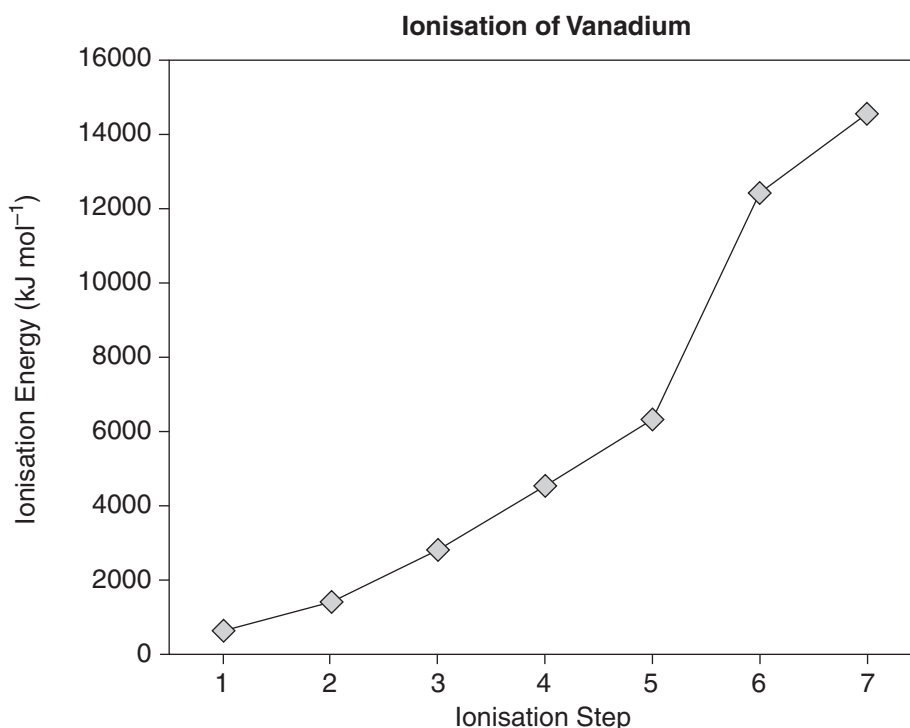


- (i) What is the oxidation state of manganese in $[\text{Mn}(\text{NH}_3)_2(\text{H}_2\text{O})_3\text{OH}]^{2+}$? **1**
- (ii) Draw a Lewis structure of ONE of these coordination complexes, and describe the bonding in this molecule. **2**
- (b) (i) Why has copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), with its strong blue colour, not been used as a pigment? **2**
- (ii) Describe a named process that can be used to attach pigments to surfaces. **3**

Question 37 continues on page 35

Question 37 (continued)

- (c) The ionisation energies of vanadium are shown in the graph.



Explain the relationship between ionisation energy and the arrangement of electrons in the vanadium atom.

5

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) (i) Describe an experiment that could be performed to relate the colour of iron solutions to the electronic configuration of iron ions, if the supplied salt is iron(II) chloride crystals.

3

- (ii) Write the electronic configuration of Fe^{2+} and Fe^{3+} .

2

- (e) Evaluate the contribution of the Bohr model to the development of our understanding of the structure of the atom.

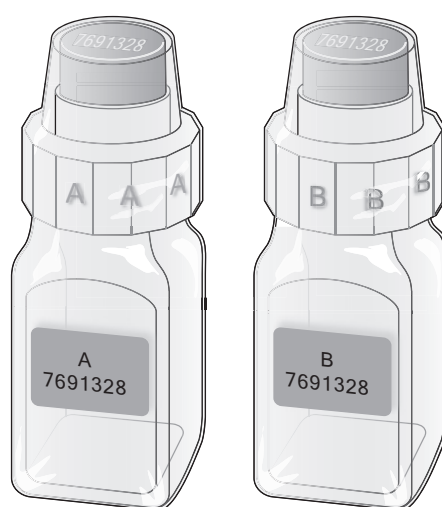
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End of Question 37

Question 38 — Forensic Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) Describe how the different characteristics of a soil sample would allow forensic chemists to trace its origin. **3**
- (b) An athlete is required to give two urine samples. Sample A is tested for evidence of substances on the banned substance list. If a banned substance is detected in Sample A then Sample B is tested. If both samples are positive for a banned substance then the athlete is asked to explain the findings at a court hearing.



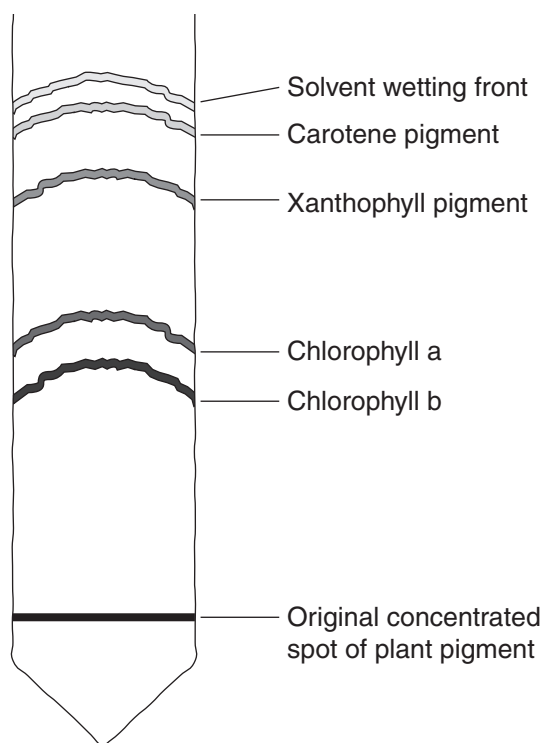
- (i) Describe what precautions should be undertaken to ensure that the evidence collected for this purpose is accepted by the court. **3**
- (ii) New technology is constantly required in forensic chemistry to keep up with drug cheats or improve the outcomes of a criminal investigation. **2**

Outline how recent advances in technology could have altered the outcome of a specific case.

Question 38 continues on page 37

Question 38 (continued)

- (c) The diagram shows the separation of plant pigments by chromatography using hexane as a solvent.



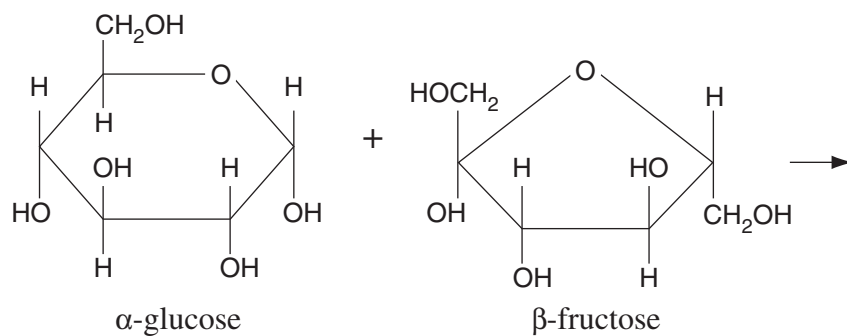
- (i) State which of the pigments is the least polar and justify your choice. **2**
- (ii) Explain how changing the solvent would affect the separation process. **3**

Question 38 continues on page 38

Question 38 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) Sucrose is a carbohydrate composed of fructose and glucose.



- | | |
|--|----------|
| (i) Complete the structural equation for the formation of sucrose. | 2 |
| (ii) Outline a chemical test that could be performed to distinguish between sucrose and the monomers fructose and glucose. Include a chemical equation in your answer. | 3 |
| | |
| (e) Analyse how modern techniques utilise the features of DNA and manipulate it to successfully generate an individual's DNA profile. | 7 |

End of paper

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol Standard Atomic Weight Name																5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon					
11 Na 22.99 Sodium		12 Mg 24.31 Magnesium																		13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon					
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon	
87 Fr Francium		88 Ra Radium		Actinoids		Rutherfordium		Dubnium		Seaborgium		Bohrium		Hassium		Meitnerium		Darmstadtium		Roentgenium		Copernicium													

Elements with atomic numbers 113 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.

2012 HSC Chemistry Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	A
4	C
5	B
6	C
7	D
8	D
9	B
10	B
11	B
12	D*
13	D
14	A
15	C
16	A
17	C
18	C
19	A
20	A

* D was the best answer for Question 12.

However, the correct answer is *2-chloro-3-fluorobutane*, which was not one of the alternatives.

Teachers are reminded that IUPAC provides a system for the clear communication of chemical nomenclature.

Section I, Part B

Question 21 (a)

Criteria	Marks
• Writes a correct balanced structural chemical equation, including catalyst	2
• Has correct structural formulae for reactants and products, may omit H ₂ O or catalyst OR • All correct but small error in a structural formula	1

Question 21 (b)

Criteria	Marks
• Identifies a precaution and provides a valid reason	2
• Identifies a precaution	1

Question 22 (a)

Criteria	Marks
• Correctly identifies the chemical process	1

Question 22 (b)

Criteria	Marks
• Gives specific reasons for the usefulness of a model	2
• Gives a reason for the usefulness of the model	1

Question 23

Criteria	Marks
- Demonstrates thorough knowledge and understanding of disturbances to pressure and temperature on the solubility of CO₂ in an equilibrium - Includes correct equation	3
- Demonstrates a sound knowledge and understanding of disturbances to pressure and temperature on the solubility of CO₂ in an equilibrium OR - Demonstrates thorough knowledge and understanding of the solubility of CO₂ in an equilibrium	2
Demonstrates a sound knowledge and understanding of the solubility of CO₂ in an equilibrium	1

Question 24

Criteria	Marks
Demonstrates a thorough knowledge of the importance of ammonia as a raw material	3
Demonstrates a sound knowledge of the importance of ammonia as a raw material	2
Demonstrates a basic knowledge of the importance of ammonia as a raw material	1

Question 25

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the process of monitoring of eutrophication in waterways - Describes tests which could be performed	3
- Demonstrates a sound knowledge and understanding of the process of monitoring of eutrophication in waterways - Gives details of the tests which could be performed	2
- Demonstrates some knowledge and understanding of the monitoring of eutrophication in waterways OR - Gives basic details of the tests which could be performed	1

Question 26 (a)

Criteria	Marks
• Demonstrates a thorough knowledge of both processes • Communicates information in a logical progression of ideas using a flow chart	5
• Demonstrates a sound knowledge of both processes • Communicates information in a logical progression of ideas using a flow chart	4
• Demonstrates a basic knowledge of both processes OR demonstrates a thorough knowledge of ONE process • Communicates information in a logical progression of ideas using a flow chart	3
• Demonstrates a limited knowledge of both processes OR • Demonstrates a basic knowledge of ONE process	2
• Demonstrates a limited knowledge of ONE process	1

Question 26 (b)

Criteria	Marks
Demonstrates a thorough knowledge of the sustainability of ethanol production from a comparison of the two sources	3
Demonstrates a sound knowledge of the sustainability of ethanol production from a comparison of the two sources	2
Demonstrates a basic knowledge of the sustainability of ethanol production from the two sources	1

Question 27 (a)

Criteria	Marks
Justifies the different uses of the radioisotopes, by demonstrating an understanding of penetration and energy transfer for both types of emission	2
Justifies the different uses of the radioisotopes, by demonstrating an understanding of penetration or energy transfer for ONE type of emission or a partial understanding of both types	1

Question 27 (b)

Criteria	Marks
Correct equation is given	1

Question 28

Criteria	Marks
Correctly calculates pH of resultant solution	3
Completes calculation with ONE error	2
- Supplies balanced chemical equation OR - Calculates moles of H_3O^+ or OH^- initially OR - Correctly performs a pH calculation	1

Question 29

Criteria	Marks
• Thorough understanding of structure of atmosphere, position and source of pollutants	4–5
• Sound understanding of structure of atmosphere, position and source of pollutants	3
• Basic understanding of structure of atmosphere, position and source of pollutants	2
• Limited understanding of structure of atmosphere, position and source of pollutants	1

Question 30 (a)

Criteria	Marks
• Correctly calculates NaOH molarity	2
• Calculates [NaOH] with ONE error OR • Calculates moles of HCl or NaOH	1

Question 30 (b) (i)

Criteria	Marks
• Correctly calculates the mass of aspirin (average)	3
• Calculates mass of aspirin with ONE error OR • Calculates moles of NaOH/C ₉ H ₈ O ₄ and molar mass aspirin	2
• Calculates moles of NaOH/C ₉ H ₈ O ₄ OR • Calculates molar mass aspirin (C ₉ H ₈ O ₄)	1

Question 30 (b) (ii)

Criteria	Marks
• Correctly identifies the solvent role of ethanol	1

Question 31 (a)

Criteria	Marks
• Correct axes/labels. Data correctly plotted	3
• Data correctly plotted without gap for butan-1-ol	2
• Data correctly plotted without gap. No labels	1

Question 31 (b)

Criteria	Marks
• Correctly predicts the boiling point from graph	1

Question 31 (c)

Criteria	Marks
• Correctly relates increasing dispersion forces to increasing boiling points	1

Question 32

Criteria	Marks
- Identifies correct concentration from calibration curve - Calculates correct concentration with correct units - Makes correct conclusion	3
- Identifies correct concentration from calibration curve and performs partial calculation and provides relevant conclusion OR - Performs calculation on incorrect concentration and provides relevant conclusion	2
- Uses calibration curve correctly and provides relevant conclusion OR - Gives partial calculation	1

Question 33

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the identified roles of chemists and the environmental impact of a correctly named electrochemical cell • Assesses the need for collaboration between chemists • Demonstrates coherence and logical progression of ideas with correct scientific terminology used	6
• Demonstrates a sound knowledge and understanding of the identified roles of chemists and the environmental impact of a correctly named electrochemical cell • Discusses collaboration between chemists • Communicates some scientific principles and ideas clearly	4–5
• Demonstrates a basic knowledge and understanding of the role of chemists and the environmental impact of a named electrochemical cell • Communicates ideas in a basic form using general scientific language	2–3
• Demonstrates a limited knowledge and understanding of the role of chemists OR the environmental impact of a named electrochemical cell • Communicates simple ideas	1

Section II

Question 34 (a)

Criteria	Marks
- Correctly identifies electrolysis - Correctly identifies gases at correct electrodes	3
- Correctly identifies electrolysis - Correctly identifies gases at incorrect electrodes OR - Correctly identifies electrolysis - Correctly identifies one gas at correct electrode OR - Correctly identifies two gases at correct electrodes	2
- Correctly identifies electrolysis OR - Correctly identifies one gas	1

Question 34 (b) (i)

Criteria	Marks
Correctly writes equation from equilibrium constant expression	1

Question 34 (b) (ii)

Criteria	Marks
• Correctly calculates K and infers position of equilibrium	3
• Calculation of K with ONE error and infers position of equilibrium OR • Correctly calculates K	2
• Identifies equilibrium concentration of one species other than N ₂ OR • Substitutes calculated data into expression for K OR • Correctly infers position of equilibrium from incorrect value of K	1

Question 34 (b) (iii)

Criteria	Marks
• Gives correct answer	1

Question 34 (c) (i)

Criteria	Marks
• Description of the process including correct equations for the production of oleum and sulfuric acid	3
• Description of the process and one correct equation for the production of oleum or sulfuric acid OR • No description and two correct equations OR • Description only and one correct equation	2
• Description only OR • One correct equation OR • Two balanced equations, no states	1

Question 34 (c) (ii)

Criteria	Marks
- States direct reaction is exothermic - Identifies difficulty in collecting $\text{H}_2\text{SO}_4(l)$	2
- States direct reaction is exothermic OR - Identifies difficulty in collecting $\text{H}_2\text{SO}_4(l)$	1

Question 34 (d) (i)

Criteria	Marks
- Description of chemical step, linked to Solvay process - Appropriate equation included	3
- Description of chemical step not linked to Solvay process - Appropriate equation included OR - Description of chemical step, linked to the Solvay process – no equation	2
- Description of chemical step only OR - Chemical equation only	1

Question 34 (d) (ii)

Criteria	Marks
Identifies one risk factor and one relevant difficulty in modelling this step	2
- Identifies one risk factor OR - Identifies one relevant difficulty in modelling this step	1

Question 34 (e)

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the relationship between structure and uses of soaps and detergents • Accounts for the development of modern detergents in terms of supply of raw materials, effectiveness in hard water, environmental concerns • Uses correct and relevant chemistry • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of relationship between structure and uses of soap and detergents • Outlines the development of modern detergents in terms of some aspects above • Uses some correct and relevant chemistry • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge and understanding of soaps and detergents and their uses • Identifies some factors relating to the development of detergents • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge and understanding of soaps and/or detergents and their uses and/or structures • Communicates simple ideas	1

Question 35 (a)

Criteria	Marks
Demonstrates a thorough understanding of the processes occurring when a saturated solution evaporates and the damage this causes	3
Demonstrates a sound understanding of the processes occurring when a saturated solution evaporates	2
- Identifies that drying the artefact could cause damage OR - Displays a limited understanding of the processes occurring when a saturated solution evaporates	1

Question 35 (b) (i)

Criteria	Marks
- Demonstrates coherence and logical progression of ideas - Demonstrates a thorough knowledge of cathodic protection by the use of applied current	3
- Communicates ideas in a clear manner - Sound understanding of cathodic protection without identification of 'impressed current' OR - Impressed current identified but explanation lacks coherence	2
- Limited understanding of cathodic protection OR - Simple statement of rust prevention	1

Question 35 (b) (ii)

Criteria	Marks
Uses correct half-equations to describe the actions of a sacrificial anode	2
Identifies use of a sacrificial anode without supporting equations	1

Question 35 (c) (i)

Criteria	Marks
- Correctly identifies locations as deep sea environments - Provides a factor for this process	2
- Correctly identifies locations as deep sea environments OR - Provides a factor for this process	1

Question 35 (c) (ii)

Criteria	Marks
- Correctly balanced equation given - Mass of iron correctly calculated with all working shown	3
- Overall equation attempted, but incorrectly balanced - Calculations correct based on equation shown OR - Overall equation correct - Calculations attempted with minimal error OR - Half-equation present and working shown for calculation of mass - Calculations contain errors in stoichiometry only	2
- Moles of H_2S calculated OR - Provides balanced equation	1

Question 35 (d) (i)

Criteria	Marks
Gives reason relating variables to validity	1

Question 35 (d) (ii)

Criteria	Marks
- Correctly draws beakers to illustrate valid investigation of TWO factors - Lists expected results for BOTH factors	4
- Correctly draws beakers to illustrate valid investigation of TWO factors - Lists one expected result OR - Draws beakers to illustrate investigation with minor error and - Lists expected results for BOTH factors	3
- Draws beakers only OR - Draws beakers and labels results for ONE investigation OR - Lists expected results for TWO factors	2
- Describes a valid investigation OR - Lists one expected result for an identified experiment	1

Question 35 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the history of ocean-going vessels, the composition of types of steel and methods of preventing corrosion • Provides a judgment about the use of steel for ship building in light of these concepts • Writes logically, providing a coherent progression of ideas	6–7
• Demonstrates a sound knowledge and understanding of the history of ocean-going vessels, the composition of types of steel and methods of preventing corrosion • Identifies benefits of using steel in ship building • Communicates ideas in a clear manner	4–5
• Demonstrates a thorough knowledge and understanding of EITHER the history of ocean-going vessels OR the composition of steel OR methods of preventing corrosion OR • Demonstrates a basic knowledge and understanding of the history of ocean-going vessels, the composition of types of steel and methods of preventing corrosion • Communicates ideas using general scientific terms	2–3
• Demonstrates a limited understanding of the history of ship building OR • Demonstrates a limited knowledge of the composition of steel OR methods to prevent corrosion	1

Question 36 (a)

Criteria	Marks
• Identifies all THREE components correctly	3
• Identifies TWO components correctly OR • Names THREE components	2
• Identifies ONE component	1

Question 36 (b) (i)

Criteria	Marks
• Provides correct formula	1

Question 36 (b) (ii)

Criteria	Marks
• Provides correct formula	1

Question 36 (b) (iii)

Criteria	Marks
• Demonstrates a sound knowledge of the THREE types of compounds	3
• Demonstrates a sound knowledge of TWO types of compounds OR • Demonstrates a basic knowledge of all THREE types of compounds	2
• Demonstrates a basic knowledge of TWO types of compounds OR • Demonstrates a limited knowledge of all THREE types of compounds	1

Question 36 (c) (i)

Criteria	Marks
Recognises relative level of oxidation	1

Question 36 (c) (ii)

Criteria	Marks
- Shows flow diagram complete with at least the eight components - Shows CO₂ from Krebs cycle - Shows ATP from Electron transport chain	4
- Shows correct order, majority of components given - Shows either CO₂ from Krebs or ATP from Electron transport chain	3
Shows some elements of flow diagram in correct order	2
Shows some elements of flow diagram	1

Question 36 (d) (i)

Criteria	Marks
- Identifies bond type in primary structure - Identifies forces and bonds that determine secondary and tertiary structure	3
- Identifies bond type in primary structure - Identifies TWO factors or bonds that determine secondary and tertiary structure	2
- Identifies bond type in primary structure OR - Identifies TWO factors or bonds that determine secondary and tertiary structure	1

Question 36 (d) (ii)

Criteria	Marks
Shows how secondary structure is denatured by pH change	2
Identifies secondary structure denatured	1

Question 36 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the biochemical pathways releasing energy during different forms of exercise • Demonstrates coherence and logical progression of scientific principles and ideas	6–7
• Demonstrates a sound knowledge and understanding of the biochemical pathways releasing energy during exercise • Communicates some scientific ideas in a clear manner	4–5
• Demonstrates a basic knowledge and understanding of the biochemical pathways releasing energy during exercise • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of the biochemical pathways releasing energy • Communicates simple ideas	1

Question 37 (a) (i)

Criteria	Marks
Provides correct oxidation state	1

Question 37 (a) (ii)

Criteria	Marks
- Structure is drawn correctly, including charge if present. Ligands show appropriate orientation (ie not bonded through hydrogen) - Explanation of bonding is correct in essential features	2
Either structure is drawn correctly or explanation of bonding is correct in essential features	1

Question 37 (b) (i)

Criteria	Marks
A correct reason is given, with both sufficient description of the problem and the conditions under which it would occur to show knowledge of the chemistry of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.	2
A simple description of the potential problem is given without reference to the chemistry of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.	1

Question 37 (b) (ii)

Criteria	Marks
Demonstrates a thorough knowledge of the process, such as naming the process, naming the binder and describing the method by which the pigment is fixed to the substrate.	3
Demonstrates a basic knowledge of the process, such as naming the process and binding agent.	2
Demonstrates limited knowledge, such as the name of a process.	1

Question 37 (c)

Criteria	Marks
Demonstrates a thorough knowledge of the topic, and presents all relevant factors, including electron-electron repulsion and inner shell stability vs valence shell stability	5
Demonstrates a sound knowledge of topic, and presents a majority of relevant factors	4
Demonstrates a basic knowledge of the topic, and presents one factor	3
Demonstrates a limited knowledge of the topic	1–2

Question 37 (d) (i)

Criteria	Marks
States: need to dissolve salt, need to oxidise Fe^{2+}, importance of using a non-contributing oxidising agent	3
States two significant factors of the experiment design	2
States one significant factor of the experiment design	1

Question 37 (d) (ii)

Criteria	Marks
Both configurations correct	2
One configuration correct or two partially correct	1

Question 37 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the place of the Bohr model of the atom including both positive and negative aspects of the model • Demonstrates coherent and logical progression of scientific principles and ideas	6–7
• Demonstrates a sound knowledge and understanding of the place of the Bohr model of the atom, including positive and/or negative aspects of the model • Communicates some scientific ideas in a clear manner	4–5
• Demonstrates a basic knowledge and understanding of the place of the Bohr model of the atom, with limited reference to positive and negative aspects of the model • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge and understanding of the place of the Bohr model of the atom, and does not address positive and negative aspects of the model • Communicates simple ideas	1

Question 38 (a)

Criteria	Marks
- Demonstrates a thorough knowledge of the types of inorganic characteristics present in soil - Links characteristics to the origins of the sample	3
- Demonstrates a sound knowledge of the types of characteristics in soil OR - Gives a list of various characteristics and links them to the origins of the samples	2
Identifies some characteristics of soil	1

Question 38 (b) (i)

Criteria	Marks
- Identifies how accuracy and reliability make the sample admissible in court - Demonstrates a thorough knowledge of precautions/ procedures necessary in a forensic investigation and links them directly to the urine samples/ sample bottles - Relates procedures to accuracy and reliability	3
- Demonstrates a sound knowledge of procedures/precautions in a forensic investigation and links them to urine testing bottles OR - Identifies precautions procedures and relates them to accuracy or reliability required for admission in court	2
- Identifies precautions or procedures OR - Identifies a precaution/procedure and relates it to validity or reliability	1

Question 38 (b) (ii)

Criteria	Marks
- Gives details of a recent case - Shows how a change in technology could advance/did change the verdict	2
Shows how changes in technology can change the outcome of a forensic investigation	1

Question 38 (c) (i)

Criteria	Marks
- Identifies correct pigment - Gives a valid justification	2
- Identifies correct pigment OR - Gives a valid justification	1

Question 38 (c) (ii)

Criteria	Marks
Demonstrates a thorough understanding of the relationship between the solvent and the separation	3
Demonstrates a sound understanding of the relationship between the solvent and the separation	2
Describes how a separation could be changed	1

Question 38 (d) (i)

Criteria	Marks
Gives correct structural formula for all products	2
- Gives correct structural equation but omits water OR - Gives structural formula with a small error; includes water	1

Question 38 (d) (ii)

Criteria	Marks
- Demonstrates a through knowledge of an appropriate chemical test - Includes an equation - Identifies results for specified sugars	3
Demonstrates a sound knowledge of an appropriate chemical test; May state results for reducing sugars only	2
Identifies a suitable reagent	1

Question 38 (e)

Criteria	Marks
• Demonstrates a clear and thorough understanding of the features of DNA that relate to forensic science • Provides a clear description of how the DNA molecule is manipulated • The molecular properties/features of DNA are linked to each manipulation involved in obtaining DNA profile • Describes how non-coding DNA varies from person to person • Demonstrates coherence and logical progression of ideas in a scientific manner • Describes how DNA profiles are unique	6–7
• Demonstrates a sound understanding of the features of DNA that relate to forensic science • Some steps in the process of obtaining a DNA profile are described • Demonstrates a sound understanding of most factors linking the molecular properties/features of DNA to a step involved in obtaining a DNA profile • States how a DNA profile allows for the identification of an individual	4–5
• Demonstrates basic understanding of the features of DNA that relate to forensic science • A step in the process of obtaining a DNA profile is described • States how a DNA profile allows for the identification of a person OR • Shows sound understanding of some factors linking the molecular properties/features of DNA to a step involved in obtaining a DNA profile	2–3
• Demonstrates a limited understanding of the features of DNA OR • A step in the process of obtaining a DNA profile is outlined OR • States how a DNA profile allows for the identification of a person	1

Chemistry

2012 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.4.5.2.1	H13
2	1	9.2.1.2.7	H9
3	1	9.4.2.2.8	H8
4	1	9.3.4.3.3	H11
5	1	9.2.3.2.56	H9, H10
6	1	9.2.5.2.3	H6, H13
7	1	9.3.1.2.2	H13, H14
8	1	9.3.4.2.9	H8
9	1	9.4.4.2.5	H6
10	1	9.4.3.3.1, 9.4.3.2.1, 9.4.3.3.4	H12, H14
11	1	9.3.3.2.6	H8
12	1	9.4.4.2.9	H10
13	1	9.2.4.2.3	H8
14	1	9.2.4.3.4	H10
15	1	9.3.2.2.1, 9.3.2.2.2	H6
16	1	9.3.2.2.4	H12
17	1	9.2.3.2.7, 9.2.3.3.6	H10, H12
18	1	9.3.3.2.6, 9.3.3.2.1, 9.3.3.3.4	H12
19	1	9.3.4.3.3	H12
20	1	9.4.5.3.1/2	H12

Section I Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.3.5.2.4	H10, H13
21 (b)	2	9.3.5.3.1	H11
22 (a)	1	9.2.1.3.3, 9.2.1.2.6	H10
22 (b)	2	9.2.1.3.3	H2, H14
23	3	9.3.2.2.5, 9.3.2.2.4	H8, H10
24	3	9.4.2.2.1	H4
25	3	9.4.5.3.2	H8
26 (a)	5	9.2.1.2.2, 9.2.3.3.2	H10, H7
26 (b)	3	9.2.2.2.1, 9.2.3.2.8	H4
27 (a)	2	9.2.5.2.6	H12
27 (b)	1	9.2.5.2.6, 9.2.1.3.1	H10, H13
28	3	9.3.3.2.5, 9.3.3.3.7	H10, H12, H13

Question	Marks	Content	Syllabus outcomes
29	5	9.4.4.2.1, 9.4.4.2.2	H13
30 (a)	2	9.3.4.3.3	H10, H12
30 (b) (i)	3	9.3.4.3.3	H10, H12
30 (b) (ii)	1	9.2.3.2.3	H8
31 (a)	3	9.2.3.2.9, 9.3.5.2.3	H13
31 (b)	1	9.3.5.2.3	H12
31 (c)	1	9.3.5.2.3	H8
32	3	9.4.3.3.5	H12, H14
33	6	9.2.4.3.3, 9.4.1.2.2	H3, H4, H7, H8, H12

Section II

Question	Marks	Content	Syllabus outcomes
Question 34		Industrial Chemistry	
(a)	3	9.5.4.3.1	H7
(b) (i)	1	9.5.2.2.2	H10, H12
(b) (ii)	3	9.5.2.3.3	H10, H12
(b) (iii)	1	9.5.2.2.3	H8
(c) (i)	3	9.5.3.2.3	H10, H14
(c) (ii)	2	9.5.3.2.3, 9.5.3.2.7	H7, H8
(d) (i)	3	9.5.6.3.1	H8, H10, H11
(d) (ii)	2	9.5.6.3.1	H12, H11
(e)	7	9.5.5.2.1, 9.5.5.3.5, 9.5.5.2.5, 9.5.5.2.3, 9.5.5.2.6	H3, H4, H9, H13
Question 35		Shipwrecks, Corrosion and Conservation	
(a)	3	9.6.7.2.2	H8
(b) (i)	3	9.6.4.2.3	H8
(b) (ii)	2	9.6.4.2.3, 9.6.4.3.4	H3, H8
(c) (i)	2	9.6.6.2.1, 9.6.6.2.2	H8
(c) (ii)	3	9.6.6.2.2, 9.6.6.3.1	H8, H10
(d) (i)	1	9.6.3.3.1	H11
(d) (ii)	4	9.6.3.2.2, 9.6.3.3.1	H11, H13
(e)	7	9.6.2.2.3, 9.6.2.3.3, 9.6.4.2.1, 9.6.4.3.1	H3, H4, H8, H13

Question 36		The Biochemistry of movement	
(a)	3	9.7.5.2.2, 9.7.5.2.3	H6
(b) (i)	1	9.7.3.2.1	H9
(b) (ii)	1	9.7.3.2.4	H9
(b) (iii)	3	9.7.3.2.2/3, 9.7.3.3.2	H9, H8
(c) (i)	1	9.7.3.2.6, 9.7.7.2.2	H7, H9
(c) (ii)	4	9.7.8.3.1, 9.7.6.2.2/3	H7, H9, H13
(d) (i)	3	9.7.4.2.3/4, 9.7.4.2.5	H6, H9
(d) (ii)	2	9.7.4.3.2	H6, H9
(e)	7	9.7.7.3.1, 9.7.8.2.2, 9.7.10.2.1	H4, H7, H8, H9
Question 37		The Chemistry of Art	
(a) (i)	1	9.8.4.2.3	H6
(a) (ii)	2	9.8.5.2.2/3/4, 9.8.5.3.1	H6, H13
(b) (i)	2	9.8.1.2.2	H8
(b) (ii)	3	9.8.1.2.4, 9.8.1.2.5	H8
(c)	5	9.8.3.2.8, 9.8.3.3.1	H6, H12c, H14
(d) (i)	3	9.8.2.2.4, 9.8.4.3.2	H6, H11
(d) (ii)	2	9.8.4.3.1	H6
(e)	7	9.8.2.2.5, 9.8.2.3.4	H2, H6
Question 38		Forensic Chemistry	
(a)	3	9.9.1.2.4	H14
(b) (i)	3	9.9.1.2.1	H14
(b) (ii)	2	9.9.1.2.5	H4
(c) (i)	2	9.9.3.3.4	H6
(c) (ii)	3	9.9.3.3.4	H8
(d) (i)	2	9.9.2.2.2	H9
(d) (ii)	3	9.9.2.3.1, 9.9.2.2.3	H9, H11
(e)	7	9.9.4	H4